

1.3.1 Institution integrates crosscutting issues relevant to Professional Ethics, Gender, Human Values, Environment and Sustainability in transacting the Curriculum

Chhatrapati Shivaji Institute of Technology, Durg believes in building a value-based inclusive development of student community. Many courses cover cross cutting topics related to professional ethics, gender, human values, environment and sustainable development. Students are encouraged to undergo internships as part of their study projects. The curriculum helps students to make a significant contribution to their development and the well-being of humanity. The college has made several efforts towards developing value based education to budding technocrats with a vision for promotion of human values to make them a better global citizen.

Professional Ethics:

Courses such as "Professional Ethics", "Human Resource Management", "Development of Society", "Project Management", "Constitution of India", "Network Security & Cryptography", etc, are included in the curriculum to study management and ethical practices in engineering. The course "Professional Ethics" integrates ethical principles into the workplace, highlighting the significance of human relations and values. Soft skills training enhances students' inner abilities, while the Placement Cell arranges placement activities, student development, and aptitude tests regularly to meet industry demands.

Human values:

Courses on Universal Human Values and Value Education for Youth Empowerment raise awareness on physical health, mental wellness, virtues, morals, and harmony. Yoga classes improve physical fitness and self-confidence, promoting inner peace and mindfulness. Various clubs encourage group work and leadership skills. The course 'Indian Constitution and Traditional Knowledge' educates students on policies, fundamental rights, and duties. Students learn to utilize legal solutions for societal issues. The institution organizes programs like Blood Donation Camps and cultural activities to instill human values in students and staff. The NSS unit adopts a village for community service.

Gender:

The institution is committed to promoting gender equity in all areas, with the establishment of the Women Empowerment Cell in the college aimed at supporting female faculty and students in realizing their full potential. This includes creating a welcoming environment for them in all activities such as recruitment, admissions, group assignments, and leadership roles. Responsibilities such as office bearers, class representatives, and club coordination are equally distributed between boys and girls.

Environment and Sustainability:

By taking environment and sustainability courses, individuals are equipped with the skills needed to assess and mitigate the environmental impacts of our utilization of natural resources and man-made infrastructures. The curriculum of courses such as 'Environmental Science and Engineering' encompasses various subjects, such as ecosystems, biodiversity, pollution, disaster management, environmental protection laws, natural resources, population, and public health. Students also learn about minimizing the use of harmful substances, optimizing energy consumption, and promoting the recyclability of obsolete products.



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Mandatory Courses Integrated with cross cutting issues relevant to Professional Ethics, Gender, Human values, Environment sustainability and other value framework enshrined in Sustainable Development Goals and National Education Policy – 2020 into the curriculum.

Mapping of the courses

B Tech First Year (Mandatory for all Branch)

S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech First Year P Group	A000111(015)	Physics-I				
2		A000112(014)	Mathematics-I*				
3		A000113(024)	Basic Electrical and Electronics Engg.				
4		A000114(037)	Engineering Graphics and Design	✓			
5		A000115(022)	Fundamentals of Computer	✓			✓
6		A000121(015)	Physics(Lab)	✓			
7		A000122(024)	Basic Electrical and Electronics Engg. (Lab)	✓			✓
8		A000123(022)	Fundamentals of Computer (Lab)	✓		✓	✓
9		A000124(037)	Engineering Graphics and Design (Lab)	✓			
10		A000105(046)	Value Education	✓		✓	
1	B Tech First Year Q Group	A000211(011)	Chemistry-I				
2		A000212(014)	Mathematics-II**				
3		A000213(022)	Programming for Problem Solving	✓			
4		A000214(046)	English	✓		✓	
5		A000215(020)	Basic Civil Engineering and Mechanics	✓			✓
6		A000221(011)	Chemistry(Lab)	✓			✓
7		A000222(022)	Programming for Problem Solving (Lab)	✓			
8		A000223(020)	Basic Civil Engg. & Mechanics (Lab)	✓		✓	✓
9		A000224(037)	Workshop Practice/ Manufacturing Process (Lab)	✓		✓	✓
10		A000225(046)	Language(Lab)	✓		✓	



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Department of Computer Science & Engineering

S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (COMPUTER SCIENCE & ENGINEERING) THIRD SEMESTER	B000311(014)	Mathematics-III				
2		B022312(022)	Data Structure & Algorithms				
3		B022313(022)	Principles of Programming Languages				
4		B022314(022)	Digital Electronics & Logic Design				
5		B022315(022)	Operating Systems				
6		B022321(022)	Data Structure & Algorithms Laboratory				
7		B022322(022)	Digital Electronics & Logic Design Laboratory				
8		B022323(022)	Operating Systems Laboratory (UNIX)				
9		B022324(022)	Software Laboratory (Sci Lab/ MATLAB)				
10		B000306(046)	Personality Development	✓		✓	
1	B Tech (COMPUTER SCIENCE & ENGINEERING) FOURTH SEMESTER	B022411(014)	Discrete Mathematics				
2		B022412(022)	Computer System Architecture				
3		B022413(022)	Database Management Systems				
4		B022414(022)	Object Oriented Programming (with Java)				
5		B022415(022)	Design & Analysis of Algorithms				
6		B022421(022)	Computer Organization & Architecture Laboratory				
7		B022422(022)	Object Oriented Programming Laboratory (Java)				
8		B022423(022)	Database Management Systems Laboratory				
9		B022424(022)	Virtual Laboratory				
10		B000406(046)	Environmental Science (Constitution of India)	✓		✓	
1	B Tech (COMPUTER SCIENCE & ENGINEERING) FIFTH SEMESTER	C022511(022)	Microprocessors & Interfaces				
2		C022512(022)	Computer Networks				
3		C022513(022)	Formal Languages and Automata Theory				
4		C022514(022)	Data Analytics with Python				
5		C022531(022)	Computer Graphics				
6		C022521(022)	Microprocessors & Interfaces Laboratory				
7		C022522(022)	Computer Networks Laboratory				
8		C022523(022)	Data Analytics with Python Laboratory				
9		C022524(022)	Project-I based on Summer Internship/ Industrial Training	✓			
10		C000506(020)	Environmental Studies				✓



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S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (COMPUTER SCIENCE & ENGINEERING) SIXTH SEMESTER	C022611(022)	Compiler Design				
2		C022612(022)	Software Engineering & Project Management				
3		C022613(022)	Artificial Intelligence & Expert Systems				
4		C022632(022)	Internet of Things				
5		C000619(022)	Cryptography & Network Security				
6		C022621(022)	Software Engineering & Project Management Laboratory				
7		C022622(022)	Artificial Intelligence & Expert Systems Lab				
8		C022624(022)	Internet of Things Lab				
9		C022627(022)	Android Application Laboratory				
10		C000601(046)	Technical Communication and Soft Skill	✓		✓	
1	B Tech (COMPUTER SCIENCE & ENGINEERING) SEVENTH SEMESTER	D022711(022)	Machine Learning				
2		D022712(022)	Data Mining and Warehousing				
3		D022713(022)	Internet and Web Technology				
4		D022732(022)	Cyber Security				
5		D000721(022)	Software Testing				
6		D022721(022)	Machine Learning Lab				
7		D022722(022)	Internet and Web Technology Lab				
8		D022723(022)	Project (Phase I)	✓			
9		D022724(022)	Industrial Training	✓			
10		D000701(046)	Universal Human Values 2			✓	
1	B Tech (COMPUTER SCIENCE & ENGINEERING) EIGHTH SEMESTER	D022811(022)	Cyber Law and Intellectual Property	✓			
2		D000836(022)	Supply Chain Management	✓			
3		D022832(022)	R Programming				
4		D022821(022)	Computer Vision Lab				
5		D022822(022)	R Programming Lab				
6		D022823(022)	Project (Phase II)	✓			



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Department of Artificial Intelligence & Data Science

S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B TECH (ARTIFICIAL INTELLIGENCE AND DATA SCIENCE) THIRD SEMESTER	B109311(014)	Mathematics-III				
2		B109312(022)	Data Structure & Algorithms				
3		B109313(022)	Operating Systems				
4		B109314(022)	Introduction to Python				
5		B109315(022)	Digital Electronics & Logic Design				
6		B109321(022)	Data Structure & Algorithms Laboratory				
7		B109322(022)	Operating Systems Laboratory				
8		B109323(022)	Digital Electronics & Logic Design Laboratory				
9		B109324(022)	Python Laboratory				
10		B000306(046)	Personality Development	✓			
1	B TECH (ARTIFICIAL INTELLIGENCE AND DATA SCIENCE) FOURTH SEMESTER	B109411(014)	Design & Analysis of Algorithms				
2		B109412(022)	Database Management Systems				
3		B109413(022)	Object Oriented Programming (with Java)				
4		B109414(022)	Computer System Architecture				
5		B109415(022)	Discrete Structure				
6		B109421(022)	Design & Analysis of Algorithms using Python Laboratory				
7		B109422(022)	Database Management Systems Laboratory				
8		B109423(022)	Object Oriented Programming Laboratory (Java)				
9		B109424(022)	Virtual Laboratory				
10		B000406(046)	Indian Culture and Constitution of India			✓	
1	B TECH (ARTIFICIAL INTELLIGENCE AND DATA SCIENCE) FIFTH SEMESTER	C109511(022)	Artificial Intelligence				
2		C109512(022)	Introduction to Machine Learning				
3		C109513(022)	Theory of Computation				
4		C109514(022)	Statistical Thinking for Data Science				
5		C109531(022)	Internet of Things (IoT)				
6		C109521(022)	Artificial Intelligence Laboratory				
7		C109522(022)	Machine Learning Laboratory				
8		C109523(022)	Data Analytics using R Programming Laboratory				
9		C109524(022)	Major Project Phase-I	✓			
10		C117525(22)	Environmental Science				✓



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S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B TECH (ARTIFICIAL INTELLIGENCE AND DATA SCIENCE) THIRD SEMESTER	B109311(014)	Mathematics-III				
2		B109312(022)	Data Structure & Algorithms				
3		B109313(022)	Operating Systems				
4		B109314(022)	Introduction to Python				
5		B109315(022)	Digital Electronics & Logic Design				
6		B109321(022)	Data Structure & Algorithms Laboratory				
7		B109322(022)	Operating Systems Laboratory				
8			Digital Electronics & Logic Design Laboratory				
9		B109323(022)					
10		B109324(022)	Python Laboratory				
1	B TECH (ARTIFICIAL INTELLIGENCE AND DATA SCIENCE) FOURTH SEMESTER	B000306(046)	Personality Development	✓			
2		B109411(014)	Design & Analysis of Algorithms				
3		B109412(022)	Database Management Systems				
4			Object Oriented Programming (with Java)				
5		B109413(022)					
6		B109414(022)	Computer System Architecture				
7		B109415(022)	Discrete Structure				
8		B109421(022)	Design & Analysis of Algorithms using Python Laboratory				
9		B109422(022)	Database Management Systems Laboratory				
10		B109423(022)	Object Oriented Programming Laboratory (Java)				
1	B TECH (ARTIFICIAL INTELLIGENCE AND DATA SCIENCE) FIFTH SEMESTER	B109424(022)	Virtual Laboratory				
2		B000406(046)	Indian Culture and Constitution of India			✓	
3		C109511(022)	Artificial Intelligence				
4		C109512(022)	Introduction to Machine Learning				
5		C109513(022)	Theory of Computation				
6		C109514(022)	Statistical Thinking for Data Science				
7		C109531(022)	Internet of Things (IoT)				
8		C109521(022)	Artificial Intelligence Laboratory				
9		C109522(022)	Machine Learning Laboratory				
10		C109523(022)	Data Analytics using R Programming Laboratory				
1		C109524(022)	Major Project Phase-I	✓			
2		C117525(22)	Environmental Science				✓



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Department of Information Technology

S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (Information Technology) THIRD SEMESTER	B000311(014)	Mathematics -III				
2		B033312(033)	Computer Architecture, Organization and Microprocessor				
3		B033313(033)	Object Oriented Concepts & Programming using JAVA				
4		B033314(033)	Computer Networks				
5		B033315(033)	Digital Electronics				
6		B033321(033)	Digital Electronics Lab				
7		B033322(033)	COA& Microprocessor Lab				
8		B033323(033)	Computer Networks Lab				
9		B033324(033)	Software Lab (Programming in Java)				
10		B000306(046)	Personality Development	✓			
1	B Tech (Information Technology) FOURTH SEMESTER	B033411(033)	Data Structures				
2		B033412(033)	Database Management System				
3		B033413(033)	Analog Electronic Circuits				
4		B033414(033)	Operating System				
5		B033415(033)	Internet of Things				
6		B033421(033)	Data Structures Lab				
7		B033422(033)	Database Management System Lab				
8		B033423(033)	Operating System (UNIX) Lab				
9		B033424(033)	Virtual Lab (IoT Lab)				
10		B000406(046)	Traditions, Culture & Constitution of India		✓	✓	✓
1	B Tech (Information Technology) FIFTH SEMESTER	C033511(033)	Artificial Intelligence & Machine Learning				
2		C033512(033)	Theory of Computation				
3		C033513(033)	Principles of Communication System				
4		C033514(033)	Software Engineering & Project Management				
5		C033531(033)	Design and Analysis of Algorithms				
6		C033521(033)	Artificial Intelligence & Machine Learning Lab				
7		C033522(033)	Software Engineering & Project Management Lab				
8		C033523(033)	Principles of Communication System Lab				
9		C033524(033)	Project-I based on Summer Internship/Industrial Training	✓			
10		C000506(020)	Environmental Studies				✓



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S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (Information Technology) 6TH SEMESTER	C033611(033)	Computer Graphics & Data Visualization				
2		C033612(033)	Compiler Design				
3		C033613(033)	Information Theory & Coding				
4		C033635(033)	Data Mining				
5		C000646(033)	Management Information System				
6		C033621(033)	Computer Graphics & Data Visualization Lab				
7		C033622(033)	Mobile Application Development(Android)Lab				
8		C033623(033)	Web Application Development Lab				
9		C033624(033)	Project –I(Android)				
10		C000601(046)	Technical Communication and Soft Skill	✓			
1	B Tech (Information Technology) 7th SEMESTER	D033711(033)	Cryptography and Network Security				
2		D033712(033)	Cloud Computing				
3		D033713(033)	Enterprise Resource Planning				
4		D033734(033)	Decision Support System				
5		D000742(033)	Managerial Skills	✓			
6		D033721(033)	Cryptography and Network Security Lab				
7		D033722(033)	Cloud Computing Lab				
8		D033723(033)	Project (PhaseI)	✓			
9		D033724(033)	Industrial Training	✓			
10		D000701(046)	Universal Human Values2			✓	
1	B Tech (Information Technology) 8th SEMESTER	D033811(033)	Big Data Analytics				
2		D033831(033)	Introduction to Data Science				
3		D000834(033)	Cyber Security and Information Security				
4		D033821(033)	Hadoop (Lab)				
5		D033822(033)	Simulation(Lab)				
6		D033823(033)	Project(Phase-II)	✓			



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Department of Mechanical Engineering

S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (MECHANICAL ENGINEERING) THIRD SEMESTER	B000311(014)	Mathematics-III				
2		B037312(037)	Mechanical Measurement and Metrology				
3		B037313(037)	Engineering Mechanics				
4		B037314(037)	Engineering Thermodynamics				
5		B037315(037)	Material Science				
6		B037321(037)	Computer Aided Machine Drawing Lab	✓			
7		B037322(037)	Mechanical Measurement and Metrology Lab	✓			
8		B037323(037)	Engineering Thermodynamics Lab	✓			
9		B037324(037)	Software Lab	✓			
10		B000306(046)	Personality Development	✓	✓	✓	
1	B Tech (MECHANICAL ENGINEERING) FOURTH SEMESTER	B037411(037)	Applied Thermodynamics				
2		B037412(037)	Fluid Mechanics				
3		B037413(037)	Strength of Materials				
4		B037414(037)	Manufacturing Process				
5		B037415(037)	Kinematic of Machine				
6		B037421(037)	Fluid Mechanics Lab	✓			
7		B037422(037)	Material Testing Lab	✓			
8		B037423(037)	Manufacturing Process Lab	✓			
9		B037424(037)	Virtual Lab	✓			
10		B000406(046)	India Cultural and Constitution of India	✓	✓	✓	✓
1	B Tech (MECHANICAL ENGINEERING) FIFTH SEMESTER	C037511(037)	Internal Combustion Engine				
2		C037512(037)	Solid Mechanics				
3		C037513(037)	Fluid Machines				
4		C037514(037)	Dynamics of Machine				
5		C037531(037)	Operation Research				
6		C037521(037)	Internal Combustion Engine Lab	✓			
7		C037522(037)	Dynamics of Machine Lab	✓			
8		C037523(037)	Fluid Machines Lab	✓			
9		C037524(037)	Project-I based on Summer Internship/Industrial Training	✓			
10		C000506(020)	Environmental Studies	✓			✓



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S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (MECHANICAL ENGINEERING) SIXTH SEMESTER	C037611(037)	Design of Machine Elements				
2		C037612(037)	Manufacturing Technology				
3		C037613(037)	Heat & Mass Transfer				✓
4		C037632(037)	Power Plant Engineering				✓
5		C000635(037)	Principles of Management	✓		✓	
6		C037621(037)	Design of Machine Elements Lab	✓			
7		C037622(037)	Computer Aided Modeling & Analysis Lab	✓			
8		C037623(037)	Heat & Mass Transfer Lab	✓			✓
9		C037624(037)	Virtual Lab-2	✓			
10		C000601(046)	Technical Communication and Soft Skill	✓	✓	✓	
1	B Tech (MECHANICAL ENGINEERING) SEVENTH SEMESTER	D037711(037)	Design of Transmission System				
2		D037712(037)	Refrigeration & Air- Conditioning				✓
3		D037713(037)	Automation in Manufacturing				
4		D037731(037)	Machine Tool Technology				
5		D000724(025)	Non Conventional Energy Sources				✓
6		D037721(037)	Refrigeration & Air- Conditioning Lab	✓			✓
7		D037722(037)	CIM & Automation Lab	✓			
8		D037723(037)	Project (Phase I)	✓		✓	
9		D037724(037)	Industrial Training	✓			
10		D000701(046)	Universal Human Values 2	✓	✓	✓	
1	B Tech (MECHANICAL ENGINEERING) EIGHTH SEMESTER	D037811(037)	Robotics				
2		D037831(037)	Automobile Engineering				✓
3		D000806(037)	Production & Product Management	✓			
4		D037821(037)	Robotics (Lab)	✓			
5		D037822(037)	Computer Aided Simulation & Analysis (Lab)	✓			
6		D037823(037)	Project (Phase-II)	✓		✓	



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Department of Mechatronics Engineering

S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (MECHATRONICS ENGINEERING) THIRD SEMESTER	B000311(014)	Mathematics-III				
2		B067312(067)	Mechanics of Materials				
3		B067313(067)	Electronic Devices and Digital Circuits				
4		B067314(067)	Metrology and Instrumentation				
5		B067315(067)	Engineering Materials				
6		B067321(067)	Material Testing Lab.	✓			
7		B067322(067)	Electronic Devices and Digital Circuits Lab.	✓			
8		B067323(067)	Metrology and Instrumentation Lab.	✓			
9		B067324(067)	Software Lab (Programming in C++)	✓			
10		B000306(046)	Personality Development	✓	✓	✓	
1	B Tech (MECHATRONICS ENGINEERING) FOURTH SEMESTER	B067411(067)	Kinematics of Machines				
2		B067412(067)	Microprocessor and Microcontroller				
3		B067413(067)	Modern Control Systems				
4		B067414(067)	Fluid Mechanics				
5		B067415(067)	Thermal Engineering				
6		B067421(067)	Kinematics of Machines Lab	✓			
7		B067422(067)	Microprocessor and Microcontroller Lab	✓			
8		B067423(067)	Fluid Mechanics Lab	✓			
9		B067424(067)	Virtual Lab(Computer aided design and manufacturing Lab)	✓			
10		B000406(046)	India Cultural and Constitution of India	✓	✓	✓	✓
1	B Tech (MECHATRONICS ENGINEERING) FIFTH SEMESTER	C067511(067)	Dynamics of Machines				
2		C067512(067)	Fluid Machines and Fluidics				
3		C067513(067)	Advance Microcontroller and System Design				
4		C067514(067)	Manufacturing Technology				
5		C067531(067)	Signals and Systems				
6		C067521(067)	Dynamic Of Machine lab	✓			
7		C067522(067)	Fluid Machines and Fluidics lab	✓			
8		C067523(067)	System Design Lab	✓			
9		C067524(067)	Project-I based on Summer Internship/ Industrial Training	✓			✓
10		C000506(020)	Environmental Studies	✓		✓	✓



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S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (MECHATRONICS ENGINEERING) SIXTH SEMESTER	C067611(067)	Machine Design				
2		C067612(067)	Industrial and Power Electronics				
3		C067613(067)	Digital Signal Processing and its Applications				
4		C067631(067)	Hydraulics and Pneumatics control				
5		C000638(067)	Entrepreneurship Developments	✓		✓	✓
6		C067621(067)	Computer Aided Modeling & Analysis Lab	✓			
7		C067622(067)	Digital Signal Processing and its Applications Lab	✓			
8		C067624(067)	Hydraulics and Pneumatics control Lab	✓			
9		C067623(067)	Simulation Lab	✓			
10		C000625(067)	Technical Communication and Soft Skill	✓	✓	✓	
1	B Tech (MECHATRONICS ENGINEERING) SEVENTH SEMESTER	D067711(067)	Autotronics				
2		D067712(067)	Automated Manufacturing I				
3		D067713(067)	Robotics and Machine Vision	✓			
4		D067735(067)	Electronic Instrumentation and Automation				
5		D000724(025)	Non Conventional Energy Sources				✓
6		D067721(067)	Autotronics Lab	✓			
7		D067722(067)	Robotics and Machine Vision Lab	✓			
8		D067723(067)	Project (Phase I)	✓		✓	✓
9		D067724(067)	Industrial Training	✓			
10		D000701(046)	Universal Human Values 2	✓	✓	✓	
1	B Tech (MECHATRONICS ENGINEERING) EIGHTH SEMESTER	D067811(067)	Automated Manufacturing–II				
2		D067831(067)	Production and Product Management	✓			
3		D000805(025)	Industrial Automation				
4		D067821(067)	Flexible Manufacturing System (Lab)	✓			
5		D067822(067)	Mechatronics Systems Design (Lab)	✓			
6		D067823(067)	Project (Phase–II)	✓		✓	✓



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Department of Civil Engineering

S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (CIVIL ENGINEERING) THIRD SEMESTER	B000311(014)	Mathematics III				
2		B020312(020)	Introduction to Fluid Mechanics	✓			
3		B020313(020)	Introduction to Solid Mechanics	✓			
4		B020314(020)	Plane Surveying	✓			
5		B020315(020)	Building Materials				✓
6		B020321(020)	Fluid Mechanics Lab	✓			
7		B020322(020)	Surveying lab	✓			
8		B020323(020)	Building Material lab	✓			✓
9		B020324(020)	Software lab	✓			
10		B000306(046)	Personality development	✓	✓	✓	
11	B Tech (CIVIL ENGINEERING) FOURTH SEMESTER	B020411(020)	Structural Analysis-I				
12		B020412(020)	Hydraulic Engineering				
13		B020413(020)	Surveying & Geomatics	✓			
14		B020414(020)	Building Construction	✓			✓
15		B020415(020)	Engineering Geology				✓
16		B020421(020)	Hydraulics Lab	✓			
17		B020422(020)	Surveying & Geometrics Lab	✓			
18		B020423(020)	Geology Lab	✓			✓
19		B020424(020)	Virtual Lab	✓			
20		B000406(046)	Indian Cultural and Constitution of India		✓	✓	✓
21	B Tech (CIVIL ENGINEERING) FIFTH SEMESTER	C020511(020)	Structural Engineering Design-I	✓			
22		C020512(020)	Hydrology & Water Resources Engineering	✓			✓
23		C020513(020)	Geotechnical Engineering				✓
24		C020514(020)	Transportation Engineering			✓	✓
25		C020531(020)	Structural Analysis-II				
26		C020521(020)	Structural Analysis Lab	✓			
27		C020522(020)	Transportation Engineering Lab	✓			✓
28		C020523(020)	Geotechnical Engineering Lab	✓			✓
29		C020524(020)	Project-I Based on Summer Internship/Industrial Training	✓		✓	
30		C000506(020)	Environmental Studies			✓	✓



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S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
31	B Tech (CIVIL ENGINEERING) SIXTH SEMESTER	C020611(020)	Structural Engineering Design-II	✓			
32		C020612(020)	Environmental Engineering	✓		✓	✓
33		C020613(020)	Engineering Economics, Estimating & Costing	✓			
34		C020632(020)	Concrete Technology				
35		C000615(020)	Construction Engineering & Management	✓			
36		C020621(020)	Structural Engineering Lab	✓			
37		C020622(020)	Environmental Engineering Lab			✓	✓
38		C020623(020)	Computer Aided Civil Engineering Drawing Lab				
39		C020624(020)	Concrete Technology Lab	✓			
40		C000601(046)	Technical Communication and Soft Skill	✓	✓	✓	
41	B Tech (CIVIL ENGINEERING) SEVENTH SEMESTER	D020711(020)	Structural Engineering Design-III	✓			
42		D020712(020)	Design of Hydraulic Structures	✓			✓
43		D020713(020)	Foundation Engineering	✓			✓
44		D020733(020)	Construction Equipment & Techniques	✓			✓
45		D000715(020)	Environmental Management & Legislation	✓		✓	✓
46		D020721(020)	Advanced Geotechnical Engineering Lab	✓			✓
47		D020722(020)	Advanced Environmental Engineering Lab	✓			✓
48		D020723(020)	Project (Phase-I)	✓	✓	✓	
49		D020724(020)	Industrial Training	✓	✓	✓	
50		D000701(046)	Universal Human values 2	✓	✓	✓	✓
51	B Tech (CIVIL ENGINEERING) EIGHTH SEMESTER	D020811(020)	Structural Engineering Design-IV	✓			
52		D020833(020)	Construction Management	✓			
53		D000814(020)	Air Pollution & Control Measures	✓			✓
54		D020821(020)	Structural Engineering Drawing Lab	✓			
55		D020822(020)	Computer Application in Civil Engineering Lab				
56		D020823(020)	Project (Phase-II)	✓	✓	✓	



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Department of Electrical & Electronics Engineering

S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
1	B Tech (EEE)/ THIRD SEMESTER	B000311(014)	Mathematics III				
2		B025312(025)	Electrical Circuit Analysis				
3		B025313(025)	Electrical Machines I	✓			
4		B025314(025)	Electronic Devices and Circuits				
5		B025315(025)	Renewable and Environment Energy				✓
6		B025321(025)	Electrical Machine I lab	✓			
7		B025322(025)	Electronic Devices and Circuit Lab	✓			
8		B025323(025)	Electrical Circuit Analysis lab	✓			
9		B025324(025)	Software lab	✓			
10		B000306(046)	Personality Development	✓			
11	B Tech (EEE)/ FOURTH SEMESTER	B025411(025)	Electro-Magnetic Field				
12		B025412(025)	Electrical Machines – II				
13		B025413(025)	Network Analysis & Synthesis				
14		B025414(025)	Electrical Measurements & Instrumentation	✓			
15		B025415(025)	Digital Electronics				
16		B025421(025)	Electrical Measurements & Instrumentation Lab	✓			
17		B025422(025)	Electrical Machines – II Lab	✓			
18		B025423(025)	Digital Electronics Lab	✓			
19		B025424(025)	Virtual lab	✓			
20		B000406(046)	India Cultural and Constitution of India	✓		✓	
21	B Tech (EEE)/FIFTH SEMESTER	C025511(025)	Signal & System				
22		C025512(025)	Control System				
23		C025513(025)	Electrical Power System-I				
24		C025514(025)	Micro Processor and Micro Controller				
25		C025535(025)	Testing & Commissioning of Electrical Equipment				
26		C025521(025)	Control System Lab	✓			
27		C025522(025)	Electrical Power System-I Lab	✓			
28		C025523(025)	Micro Processor and Micro Controller Lab	✓			
29		C025524(025)	Project-I based on Summer Internship/ Industrial Training				
30		C000506(020)	Environmental Studies	✓		✓	✓



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S.N.	PROGRAM Name/ SEMESTER	Course Code	Course Name	Professional Ethics	Gender Sensitisation	Human values	Environment sustainability
31	B Tech (EEE)/ SIXTH SEMESTER	C025611(025)	Power Electronics				
32		C025612(025)	Electrical Power System-II				
33		C025613(025)	Digital Signal Processing				
34		C025632(025)	Distributed Generation				
35		C000626(025)	Energy Auditing and Management				
36		C025621(025)	Power Electronics lab				
37		C025622(025)	Electrical Power System-II Lab	✓			
38		C025623(025)	Computer Simulation Lab	✓			
39		C025624(025)	Digital Signal Processing lab	✓			
40		C000601(046)	Technical Communication and Soft Skill	✓		✓	
41	B Tech (EEE)/ SEVENTH SEMESTER	D025711(025)	Power System Protection				
42		D025712(025)	Electrical Drives				
43		D025713(025)	High Voltage Engineering				
44		D025731(025)	Soft Computing Techniques				
45		D000724(025)	Non Conventional Energy Sources				
46		D025721(025)	Power System Protection Lab	✓			
47		D025722(025)	High Voltage Engineering Lab	✓			
48		D025723(025)	Project (Phase I)	✓			
49		D025724(025)	Industrial Training	✓		✓	
50		D000701(046)	Universal Human Values 2	✓		✓	
51	B Tech (EEE)/ EIGHTH SEMESTER	D025811(025)	Computer Aided Power System				
52		D025834(025)	EHV AC & DC Transmission				
53		D000823(076)	Management Concept Technique	✓		✓	
54		D025821 (025)	Computer Aided Power System Lab	✓			
55		D025822 (025)	IMTEE Lab	✓			
56		D025823 (025)	Project Phase-II	✓		✓	



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Schemes of the CSVTU

Schemes reflecting mandatory courses concerning cross cutting issues relevant to Professional Ethics, Gender, Human values, Environment sustainability and other value framework enshrined in Sustainable Development Goals and National Education Policy – 2020. These courses are common to all branches.

Summary of the Courses

S.No.	Name of the Courses	Course Code	Semester
1	Value Education	A000105(046)	I
2	Language (Lab)	A000225(046)	II
3	Personality Development	B000306(046)	III
4	Indian Culture and Constitution of India	B000406(046)	IV
5	Environmental Studies	C000506(020)	V
6	Technical Communication and Soft Skill	C000601(046)	VI
7	Universal Human Values 2	D000701(046)	VII



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**Chhattisgarh Swami Vivekanand Technical
University (CSVTU), Bhilai (CG)**

SCHEME OF TEACHING AND EXAMINATION

Courses of Study and Scheme of Examination of **P1 Group**
B.Tech (First Semester - Common to all Branches of Engineering) 2019-20

Sl. No.	Board of Studies (BOS)	Courses (Subject)	Course Code	Period per Week			Scheme of Examination			Total Marks	Credit
				L	T	P	Theory /Lab				
							ESE	CT	TA		
1.	Basic Sciences	Physics-I	A000111(015)	3	1	-	100	20	30	150	4
2.	Basic Sciences	Mathematics-I*	A000112(014)	3	1	-	100	20	30	150	4
3.	Electrical Engineering	Basic Electrical and Electronics Engg.	A000113(024)	2	1	-	100	20	30	150	3
4.	Mechanical Engineering	Engineering Graphics and Design	A000114(037)	1	0	-	100	20	30	150	1
5.	Computer Science	Fundamentals of Computer	A000115(022)	2	0	-	100	20	30	150	2
6.	Basic Sciences	Physics (Lab)	A000121(015)	-	-	2	40	-	20	60	1
7.	Electrical Engineering	Basic Electrical and Electronics Engg. (Lab)	A000122(024)	-	-	2	40	-	20	60	1
8.	Computer Science	Fundamentals of Computer (Lab)	A000123(022)	-	-	2	40	-	20	60	1
9.	Mechanical Engineering	Engineering Graphics and Design (Lab)	A000124(037)	-	-	4	40	-	20	60	2
10.	Humanities	Value Education	A000105(046)	-	-	-	-	-	10	10	-
Total Marks				11	3	10	660	100	240	1000	19

L-Lecture, T-Tutorial, P-Practical, ESE-End Semester Exam, CT- Class Test, TA-Teacher's Assessment

Note :

- (a) The teaching in the 1st and 2nd Semester will be divided in two groups consisting of various branches as shown below :

P1-GROUP: Electronics & Telecommunication, Mechanical, Civil, Mining, Applied Electronics & Instrumentation, Metallurgy, Mechatronics, Automobile, Production Engineering, Fashion and Apparel Engineering.

Q1-GROUP: Computer Science, Information Technology, Electronics & Instrumentation, Electrical, Chemical, Electrical & Electronics, Plastic Engineering, Agriculture Engineering, Biotechnology, Artificial Intelligence, Artificial Intelligence and Machine Learning, Artificial Intelligence and Data Sciences, Data Sciences, Cyber Security, Electronic and Computer Engineering, Computer Science and Technology.

- (b) *Mathematics-I will be taught to both the groups in the first semester.
(c) Value Education will be conducted by the relevant discipline/humanities as decided by the Principal.



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**Chhattisgarh Swami Vivekanand Technical
University (CSVTU), Bhilai (CG)**

SCHEME OF TEACHING AND EXAMINATION

Courses of Study and Scheme of Examination of **Q1 Group**
B.Tech (Second Semester - Common to all Branches of Engineering) 2019-20

Sl. No.	Board of Studies (BOS)	Courses (Subject)	Course Code	Period per Week			Scheme of Examination			Total Marks	Credit
				L	T	P	Theory /Lab				
							ESE	CT	TA		
1.	Basic Sciences	Chemistry-I	A000211(011)	3	1	-	100	20	30	150	4
2.	Basic Sciences	Mathematics-II**	A000212(014)	3	1	-	100	20	30	150	4
3.	Computer Science	Programming for Problem Solving	A000213(022)	3	-	-	100	20	30	150	3
4.	Humanities	English	A000214(046)	2	-	-	100	20	30	150	2
5.	Civil Engineering	Basic Civil Engineering and Mechanics	A000215(020)	3	-	-	100	20	30	150	3
6.	Basic Sciences	Chemistry (Lab)	A000221(011)	-	-	2	40	-	20	60	1
7.	Computer Science	Programming for Problem Solving (Lab)	A000222(022)	-	-	4	40	-	20	60	2
8.	Civil Engineering	Basic Civil Engg. & Mechanics (Lab)	A000223(020)	-	-	2	40	-	20	60	1
9.	Mechanical Engineering	Workshop Practice/ Manufacturing Process (Lab)	A000224(037)	-	1	4	40	-	20	60	3
10.	Humanities	Language (Lab)	A000225(046)	-	-	2	-	-	10	10	1
Total Marks				14	3	14	660	100	240	1000	24

L-Lecture, T-Tutorial, P-Practical, ESE-End Semester Exam, CT- Class Test, TA-Teacher's Assessment

Note :

- (a) The teaching in the 1st and 2nd Semester will be divided in two groups consisting of various branches as shown below :
- P1-GROUP:** Electronics & Telecommunication, Mechanical, Civil, Mining, Applied Electronics & Instrumentation, Metallurgy, Mechatronics, Automobile, Production Engineering, Fashion and Apparel Engineering.
- Q1-GROUP :** Computer Science, Information Technology, Electronics & Instrumentation, Electrical, Chemical, Electrical & Electronics, Plastic Engineering, Agriculture Engineering, Biotechnology, Artificial Intelligence, Artificial Intelligence and Machine Learning, Artificial Intelligence and Data Sciences, Data Sciences, Cyber Security, Electronic and Computer Engineering, Computer Science and Technology.
- (b) **Mathematics-II will be taught to both the groups in the second semester.



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Chhattisgarh Swami Vivekanand Technical
University (CSVTU), Bhilai (C.G.)

SCHEME OF TEACHING AND EXAMINATION

B.Tech (Third Semester – Civil Engineering)

Sl. No.	Board of Studies (BOS)	Courses (Subject)	Course Code	Period per Week			Scheme of Examination			Total Marks	Credit
				L	T	P	Theory/Lab				
							ESE	CT	TA		
1.	Basic Science	Mathematics - III	B000311(014)	3	1	-	100	20	30	150	4
2.	Civil Engg.	Introduction to Fluid Mechanics	B020312(020)	3	1	-	100	20	30	150	4
3.	Civil Engg.	Introduction to Solid Mechanics	B020313(020)	3	1	-	100	20	30	150	4
4.	Civil Engg.	Plane Surveying	B020314(020)	3		-	100	20	30	150	3
5.	Civil Engg.	Building Materials	B020315(020)	2	-	-	100	20	30	150	2
6.	Civil Engg.	Fluid Mechanics Lab	B020321(020)	-	-	2	40	-	20	60	1
7.	Civil Engg.	Surveying Lab	B020322(020)	-	-	2	40	-	20	60	1
8.	Civil Engg.	Building Material Lab	B020323(020)	-	-	2	40	-	20	60	1
9.	Civil Engg.	Software Lab	B020324(020)	-	-	2	40	-	20	60	1
10.	Humanities	Personality Development	B000306(046)	-	-	2	-	-	10	10	-
Total Marks:				14	3	10	660	100	240	1000	21

L-Lecture, T-Tutorial, P-Practical, ESE-End Semester Exam, CT- Class Test,
TA-Teacher's Assessment



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Chhattisgarh Swami Vivekanand Technical University (CSVTU), Bhilai (CG)

SCHEME OF TEACHING AND EXAMINATION

Courses of Study and Scheme of Examination

B. Tech. (4th Semester) –Civil Engineering

Sl. No.	Board of Studies (BOS)	Courses (Subject)	Course Code	Period per Week			Scheme of Examination			Total Marks	Credit
				L	T	P	Theory/Lab				
							ESE	CT	TA		
1.	Civil Engg.	Structural Analysis-I	B020411(020)	3	1	-	100	20	30	150	4
2.	Civil Engg.	Hydraulic Engineering	B020412(020)	3	1	-	100	20	30	150	4
3.	Civil Engg.	Surveying and geomatics	B020413(020)	3	1	-	100	20	30	150	4
4.	Civil Engg.	Building Construction	B020414(020)	3		-	100	20	30	150	3
5.	Civil Engg.	Engineering Geology	B020415(020)	2		-	100	20	30	150	2
6.	Civil Engg.	Hydraulics Lab	B020421(020)	-	-	2	40	-	20	60	1
7.	Civil Engg.	Surveying and Geomatics Lab	B020422(020)	-	-	2	40	-	20	60	1
8.	Civil Engg.	Geology Lab	B020423(020)	-	-	2	40	-	20	60	1
9.	Civil Engg.	Virtual Lab	B020424(020)	-	-	2	40	-	20	60	1
10.	Non- credit course	Indian Culture and Constitution of India	B000406(046)	-	-	2	-	-	10	10	-
Total Marks:				14	3	10	660	100	240	1000	21
Internship-I to be completed after fourth semester exam and its evaluation to be done in fifth semester											

L-Lecture, T-Tutorial, P-Practical, ESE-End Semester Exam, CT- Class Test, TA- Teacher's Assessment



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Chhattisgarh Swami Vivekanand Technical
University (CSVТУ, NEWAI (C.G.))

SCHEME OF TEACHING AND EXAMINATION

B Tech (Fifth Semester - CIVIL ENGINEERING)

Sl. No.	Board of Studies	Courses (Subject)	Course Code	Period per Week			Theory/Lab			Total Marks	Credits
				L	T	P	ESE	CT	TA		
1.	Civil Engg.	Structural Engineering Design-I	C020511(020)	3	1	-	100	20	30	150	4
2.	Civil Engg.	Hydrology & Water Resources Engineering	C020512(020)	3	1	-	100	20	30	150	4
3.	Civil Engg.	Geotechnical Engineering	C020513(020)	3	1	-	100	20	30	150	4
4.	Civil Engg.	Transportation Engineering	C020514(020)	2	1	-	100	20	30	150	3
5.	Professional Elective -I(Refer Table I)			2	0	-	100	20	30	150	2
6.	Civil Engg.	Structural Analysis lab	C020521(020)	-	-	2	40	-	20	60	1
7.	Civil Engg.	Transportation Engineering Lab	C020522(020)	-	-	2	40	-	20	60	1
8.	Civil Engg.	Geotechnical Engineering Lab	C020523(020)	-	-	2	40	-	20	60	1
9.	Civil Engg.	Project-I based on Summer Internship/ Industrial Training	C020524(020)	-	-	2	40	-	20	60	1
10.	Civil Engg.	Environmental Studies	C000506(020)	-	-	2	-	-	10	10	-
Total				13	4	10	660	100	240	1000	21

L - Lecturer

P - Practical

TA - Teacher's Assessment

T - Tutorial

ESE - End Semester Exam

CT - Class Test

Professional Elective I

S.N.	Board of Studies	Course Code	Subject
1	Civil Engg	C020531(020)	Structural Analysis-II
2	Civil Engg	C020532(020)	Repairs and rehabilitation of structures
3	Civil Engg	C020533(020)	Groundwater Engineering

Note: (1) 1/4th of total strength of students subject to minimum of 20 students is required to offer and elective in the college in a particular academic session.

(2) Choice of elective course once made for an examination cannot be changed in future examinations.



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Chhattisgarh Swami Vivekanand Technical University



(CSVTU, NEWAI (C.G.))

SCHEME OF TEACHING AND EXAMINATION

B Tech (Sixth Semester – Civil Engineering)

Sl. No.	Board of Studies	Courses (Subject)	Course Code	Period per Week			Theory/Lab			Total Marks	Credits
				L	T	P	ESE	CT	TA		
1.	Civil Engg	Structural Engineering Design-II	C020611(020)	3	1	-	100	20	30	150	4
2.	Civil Engg	Environmental Engineering	C020612(020)	3	1	-	100	20	30	150	4
3.	Civil Engg	Engineering economics, estimating and costing	C020613(020)	3	1	-	100	20	30	150	4
4.	Professional Elective-II (Refer Table I)			2	1	-	100	20	30	150	3
5.	Open Elective – I (Refer Table II)			2	0	-	100	20	30	150	2
6.	Civil Engg	Structural Engineering Lab	C020621(020)	-	-	2	40	-	20	60	1
7.	Civil Engg	Environmental Engineering Lab	C020622(020)	-	-	2	40	-	20	60	1
8.	Civil Engg	Computer Aided Civil Engineering Drawing lab	C020623(020)	-	-	2	40	-	20	60	1
9.	Civil Engg	Concrete Technology Lab	C020624(020)	-	-	2	40	-	20	60	1
10.	Humanities	Technical Communication and Soft Skill	C000601(046)	-	-	2	-	-	10	10	-
Total				13	4	10	660	100	240	1000	21

L – Lecturer
P – Practical,

T – Tutorial, TA – Teacher's Assessment
ESE – End Semester Exam, CT – Class Test

Table I (Professional Elective II)

S.N.	Board of Studies	Subject	Course Code
1.	Civil Engg	Structural Analysis by matrix method	C020631(020)
2.	Civil Engg	Concrete technology	C020632(020)
3.	Civil Engg	Traffic engineering and management	C020633(020)

Note: (1) 1/4th of total strength of students subject to minimum of 20 students is required to offer and elective in the college in a particular academic session.

(2) Choice of elective course once made for an examination cannot be changed in future examinations.



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Chhattisgarh Swami Vivekanand Technical
University (CSVТУ, NEWAI (C.G.))

SCHEME OF TEACHING AND EXAMINATION

B Tech (Seventh Semester - Civil Engineering)

Sl No.	(BOS)	Courses (Subject)	Course Code	Period per Week			Scheme of Examination			Total Marks	Credit
				L	T	P	Theory/Lab				
							ESE	CT	TA		
1.	Civil Engrg	Structural design-III	D020711(020)	3	1	-	100	20	30	150	4
2.	Civil Engrg	Design of hydraulic structures	D020712(020)	2	1	-	100	20	30	150	3
3.	Civil Engrg	Foundation Engineering	D020713(020)	2	1	-	100	20	30	150	3
4.	Professional Elective -III (Refer to Table I)			1	1	-	100	20	30	150	2
5.	Open Elective- II (Refer to Table II)			2	0	-	100	20	30	150	2
6.	Civil Engrg	Advanced Geotechnical Engineering lab.	D020721(020)	-	-	2	40	-	20	60	1
7.	Civil Engrg	Advanced Environmental engineering lab	D020722(020)	-	-	2	40	-	20	60	1
8.	Civil Engrg	Project (Phase I)	D020723(020)	-	-	6	60	-	40	100	3
9.	Civil Engrg	Industrial Training	D020724(020)			2			20	20	1
10.	Humanities	Universal Human Values 2	D000701(046)	-	-	2	-	-	10	10	-
Total Marks:				10	4	14	640	100	260	1000	20

L - Lecturer, T - Tutorial, P - Practical, CT - Class Test ESE - End Semester Exam TA - Teacher's Assessment

➤ Note: - The students have to attend the four weeks industrial training / summer internship in B. Tech program after sixth semester, which will be evaluated in seventh semester.

Table I (Professional Elective III)

S.N.	Board of Studies	Course Code	Subject
1.	Civil Engineering	D020731(020)	Seismic design of structures
2.	Civil Engineering	D020732(020)	Pre stressed Concrete
3.	Civil Engineering	D020733(020)	Construction equipment and techniques

Note: (1) 1/4th of total strength of students subject to minimum of 20 students is required to offer and elective in the college in a particular academic session.

(2) Choice of elective course once made for an examination cannot be changed in future examinations.



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Syllabus of the CSVTU

Schemes reflecting mandatory courses concerning cross cutting issues relevant to Professional Ethics, Gender, Human values, Environment sustainability and other value framework enshrined in Sustainable Development Goals and National Education Policy – 2020. These courses are common to all branches.

Chhattisgarh Swami Vivekanand Technical University, Bhilai

Name of program: Bachelor of Technology

Branch: Common to All Branches

Subject: Indian Culture and Constitution of India

Total Theory Periods: 2/Week

Assignments: Two (Minimum)

Total Marks in ESE: NIL

Semester: IV

Code: B000406(046)

Total Tutorial Periods: NIL

Marks in TA: 10

Objective: The Constitution is the supreme law and it helps to maintain integrity in the society and to promote unity among the citizens to build a great nation. The main objective of the Indian Constitution is to promote harmony throughout the nation.

Course Objectives

Upon completion of this course, the student shall be able

- To understand Meaning and concepts of Traditional and Modern of Culture
- To understand Sources of the Study of Indian Culture
- To Enable the student to understand the history and importance of constitution
- To understand philosophy of fundamental rights and duties
- To understand the powers and functions of executive, legislature and judiciary
- To understand the powers and functions of state government
- To understand the recent trends in Indian constitutional and election commission of India.

UNIT-I

Meaning and concepts of Culture: Traditional and Modern concepts of Culture-Notions of Culture in textual tradition, anthropological, archaeological and sociological understanding of the term culture. Elements of Culture, concept of Indianness and value system. Relation between culture and civilization. Historiography and approaches to the study of Indian Culture- Stereotypes, Objectivity and Bias, Imperialist, Nationalist, Marxist and Subaltern. Heritage of India and world's debt to Indian Culture.

UNIT-II

Sources of the Study of Indian Culture: Archaeological: cultural remains, Monuments, Numismatics, Epigraphy; Literary sources and Oral traditions; Foreign Accounts; Archival sources.

UNIT-III

History of Indian Constitution Constitutional History, Preamble salient features, citizenship, Method of Amendment and Recent Amendments. **Rights and Duties** Fundamental Rights and Directive Principles of State Policy. Fundamental Duties. Difference between Fundamental Rights and Directive Principles of State Policy. **Union Government** a) President-powers and functions. Vice president powers and functions, Prime Minister and council of ministers powers and functions. b) Parliament- Lok Sabha, Rajya Sabha- composition powers and functions. c) Judiciary (Supreme Court) composition powers and functions Judicial Activism



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UNIT-IV

State Government a) Governor: powers and functions b) Chief minister: powers and functions c) State Legislative Assembly and Legislative Council- composition powers and functions. d) High Court : composition powers and functions

UNIT-V

Recent Trends in Indian Constitutional a) Basic structure of Indian Constitution. b) Electoral Reforms c) Panchayati Raj system in India.

Books of Reference

1. Dr. P. K. Agrawal Indian Culture, Art and Heritage.
2. P. Raghunadha Rao Indian Heritage and Culture
3. M.V.Pylee, An Introduction to the Constitution of India, New Delhi, Vikas, 2005.
4. Subhash C. Kashyap, Our Constitution: An Introduction to India's Constitution and constitutional Law, New Delhi, National Book Trust, 2000.
5. Durga Das Basu, Introduction to the Constitution of India, New Delhi, Prentice Hall of India, 2001.
6. D.C. Gupta, Indian Government and Politics, VIII Edition, New Delhi, Vikas, 1994.
7. V.D. Mahajan, Constitutional Development and National Movement in India, New Delhi, S. Chand and Co., latest edition.



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Chhattisgarh Swami Vivekanand Technical University, Bhilai

Name of the Program: B.Tech

Semester: V

Subject: Environmental Studies

Code: C000506(020)

Period per week (L-T-P): (0-0-2) / Week

Non-Credit

Total Contact Hours: 40

No. of assignments to be submitted: 05

PREREQUISITE: Knowledge of basic Chemistry, Physics and Mathematics.

COURSE OBJECTIVES:

1. Basic knowledge of environment, ecology, ecosystems, biodiversity and conservation.
2. Fundamentals of natural resources, control, uses and its impact on environment.
3. Human population, growth, growing needs and its impact on society and environment.
4. Types of environmental pollution, legislations, enactment and management.

COURSE DETAILS:

UNIT I: Introduction to environmental studies, ecology and ecosystems

(06 hours)

Introduction to environment; Concept and structure of ecology and ecosystem, energy flow; Community ecology; Food chains and webs; Ecological succession; Characteristic features of forest, grassland, desert and aquatic ecosystem; Multidisciplinary nature of environmental studies, scope and importance; Concept of sustainability and sustainable development.

UNIT II: Biodiversity and conservation

(06 hours)

Introduction to biological diversity and levels of genetic, species and ecosystem diversity; Biogeographic zones of India; Biodiversity patterns and global biodiversity hot spots; Threats to biodiversity, habitat loss, conflicts and biological invasions; In-situ and Ex-situ conservation of biodiversity; Ecosystem and biodiversity services.

UNIT III: Natural resources and environment

(08 hours)

Concept of Renewable and non-renewable resources; Land resources, land use change, land degradation, soil erosion; Desertification; Deforestation: causes, consequences and remedial measures; Water: Use and over-exploitation of surface and ground water, floods, droughts, conflicts over water (international & inter-state); Energy resources: environmental impacts of energy generation, use of alternative and nonconventional energy sources, growing energy needs.

UNIT IV: Human communities, social issues and environment

(08 hours)

Basic concept of human population, growth and communities; Impacts on environment, human health, welfare and human rights; Resettlement and rehabilitation; Environmental natural disaster: floods, earthquakes, cyclones, tsunami and landslides; Manmade disaster; Environmental movements; Environmental ethics: role of gender and cultures in environmental conservation; Environmental education and public awareness; Human health risks and preventive measurements.

UNIT V: Environmental pollution, policies, legislations, assessment and practices

(12 hours)

Environmental pollution: Causes, effects and controls of air, water, soil, noise and marine pollution; Concept of hazardous and non-hazardous wastes, biomedical and e-wastes; Solid waste management and control measures; Climate change, global warming, ozone layer depletion, acid rain and their societal impacts; Environment laws: Wildlife Protection Act, Forest Conservation Act, Water (Prevention and control of Pollution) Act, Air (Prevention & Control of Pollution) Act, Environment Protection Act, Biodiversity Act, International agreements negotiations, protocols and practices; EIA, EMP.



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On completion of each unit, students have to submit one assignment from each unit.

COURSE OUTCOMES (CO):

On completion of the course, students will able to:

1. Interpret and demonstrate the concept of ecology and ecosystem for environmental sustainability.
2. Define and establish the diversified knowledge of biodiversity and its conservation.
3. Explain the uses of natural resources efficiently and its impact on environment.
4. Illustrate and solve the simple and complex social issues relating to human communities.
5. Exemplify and make useful solution to combat the environmental degradation with the aid of national and international legislations and protocols there under.
6. Demonstrate and elucidate the complicated issues and anthropological problems for societal development.

TEXT BOOKS:

1. De, A.K., (2006). *Environmental Chemistry*, 6th Edition, New Age International, New Delhi.
2. Bharucha, E. (2013). *Textbook of Environmental Studies for Undergraduate Courses*. Universities Press.
3. Asthana, D. K. (2006). *Text Book of Environmental Studies*. S. Chand Publishing.

REFERENCE BOOKS:

1. Odum, E. P., Odum, H. T., & Andrews, J. (1971). *Fundamentals of ecology*. Philadelphia: Saunders.
2. Basu, M., Xavier, S. (2016). *Fundamentals of Environmental Studies*, Cambridge University Press, India.
3. Sharma, P. D., & Sharma, P. D. (2005). *Ecology and Environment*. Rastogi Publications.

OPEN SOURCE LEARNING:

<http://nptel.ac.in/>



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Chhattisgarh Swami Vivekanand Technical University, Bhilai

Program / Semester: B.Tech (VI)	Branch: Humanities
Subject: Technical Communication & Soft Skills	Course Code: C000601(046)
Total Marks (Internal Assessment): 10	L: 0 T:0 P: 2 Credit(s): 0
Internal Assessments to be conducted: 02	Duration (End Semester Exam): NA

UNIT-1 Communication Skills-Basics: Understanding the communicative environment, Verbal Communication; Non Verbal Communication & Cross Cultural Communication, Body Language & Listening Skills; Employment Communication & writing CVs, Cover Letters for correspondence. Common errors during communication, Humour in Communication.

UNIT-2 Interpersonal communication: Presentation, Interaction and Feedbacks, Stage Manners, Group Discussions (GDs) and facing Personal Interviews, Building Relationships, Understanding Group Dynamics- I, Emotional and Social Skills, Groups, Conflicts and their Resolution, Social Network, Media and Extending Our Identities.

UNIT-3 Vocational skills: Managing time: Planning and Goalsetting, managing stress: Types of Stress; Making best out of Stress, Resilience, Work-life balance, Applying soft-skills to workplace.

UNIT-4 Mindsets and Handling People: Definitions and types of Mindset, Learning Mindset, Developing Growth Mindset, Types of People, How to Lead a Meeting, How to Speak Effectively in Meetings, Behavior & Roles in Meetings, Role Play: Meeting. On Saying "Please", How to say "NO".

UNIT-5 Positive Psychology: Motivating oneself, Persuasion, Survival Strategies, Negotiation, Leadership and motivating others, controlling anger, Gaining Power from Positive Thinking.

Text Books:

1. Peter S. J., Francis. Soft Skills and Professional Communication. New Delhi: Tata McGraw-Hill Education, 2011.
2. Stein, Steven J. & Howard E. Book. The EQ Edge: Emotional Intelligence and Your Success. Canada: Wiley & Sons, 2006.
3. Dorch, Patricia. What Are Soft Skills? New York: Execu Dress Publisher, 2013.

Reference Books:

- Kamin, Maxine. Soft Skills Revolution: A Guide for Connecting with Compassion for Trainers, Teams, and Leaders. Washington, DC: Pfeiffer & Company, 2013.
- Peale Norman Vincent. The Power of Positive Thinking: 10 Traits for Maximum Result. Paperback Publication. 2011.
- Klaus, Peggy, Jane Rohman & Molly Hamaker. The Hard Truth about Soft Skills. London: Harper Collins E-books, 2007.

Course Outcomes:

1. Learn to listen actively to analyse audience and tailor the delivery accordingly.
2. Increase their awareness of communication behaviour by using propriety-profiling tool.
3. Master three "As" of stressful situation: Avoid, Alter, Accept; to cope with stressors and create a plan to reduce or eliminate them.
4. Develop growth mind-set and able to handle difficult person and situations successfully.
5. Develop technique of turning negativity into positivity and generate self-motivation skills.



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Chhattisgarh Swami Vivekananda Technical University, Newai

Name of the Program: Bachelor of Technology

Semester: B. Tech – 7th

Subject: Universal Human values 2

Total Marks in End Semester Exam:

Branch: Civil Engg.

Course Code: D000701(046)

L: T: P: 2 Credits: 0

Course Objective(s):

- Development of a holistic perspective based on self- exploration about themselves (human being), family, society and nature/existence.
- Understanding (or developing clarity) of the harmony in the human being, family, society and nature/existence
- Strengthening of self-reflection.
- Development of commitment and courage to act.

UNIT-I Introduction- Need, Basic Guidelines, Content and Process for Value Education

- Purpose and motivation for the course, recapitulation from Universal Human Values-I.
- Self-Exploration-what is it? - Its content and process; 'Natural Acceptance' and Experiential Validation- as the process for self-exploration.
- Continuous Happiness and Prosperity- A look at basic Human Aspirations
- Right understanding, Relationship and Physical Facility- the basic requirements for fulfillment of aspirations of every human being with their correct priority.
- Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario
- Method to fulfill the above human aspirations: understanding and living in harmony at various levels.
- Include practice sessions to discuss natural acceptance in human being as the innate acceptance for living with responsibility (living in relationship, harmony and co-existence) rather than as arbitrariness in choice based on liking-disliking.

UNIT-II Understanding Harmony in the Human Being - Harmony in Myself!

- Understanding the needs of Self ('I') and 'Body' - happiness and physical facility.
- Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer).
- Understanding the characteristics and activities of 'I' and harmony in 'I'.
- Understanding the harmony of I with the Body: Sanyam and Health; correct appraisal of Physical needs, meaning of Prosperity in detail.
- Programs to ensure Sanyam and Health.
- Include practice sessions to discuss the role others have played in making material goods available to me. Identifying from one's own life.
- Differentiate between prosperity and accumulation. Discuss program for ensuring health vs dealing with disease

UNIT-III Understanding Harmony in the Family and Society- Harmony in Human-Human Relationship



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- Understanding values in human-human relationship; meaning of Justice (nine universal values in relationships) and program for its fulfilment to ensure mutual happiness; Trust and Respect as the foundational values of relationship
- Understanding the meaning of Trust; Difference between intention and competence
- Understanding the meaning of Respect, Difference between respect and differentiation; the other salient values in relationship
- Understanding the harmony in the society (society being an extension of family): Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals
- Visualizing a universal harmonious order in society- Undivided Society, Universal Order- from family to world family.
- Include practice sessions to reflect on relationships in family, hostel and institute as extended family, real life examples, teacher-student relationship, goal of education etc. Gratitude as a universal value in relationships. Discuss with scenarios. Elicit examples from students' lives.

UNIT-IV Understanding Harmony in the Nature and Existence - Whole existence as Coexistence

- Understanding the harmony in the Nature
- Interconnectedness and mutual fulfilment among the four orders of nature- recyclability and self-regulation in nature.
- Understanding Existence as Co-existence of mutually interacting units in all-pervasive space.
- Holistic perception of harmony at all levels of existence.
- Include practice sessions to discuss human being as cause of imbalance in nature (film "Home" can be used), pollution, depletion of resources and role of technology etc.

UNIT-V Implications of the above Holistic Understanding of Harmony on Professional Ethics

- Natural acceptance of human values
- Definitiveness of Ethical Human Conduct
- Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order
- Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems.
- Case studies of typical holistic technologies, management models and production systems
- Strategy for transition from the present state to Universal Human Order:
 - o At the level of individual: as socially and ecologically responsible engineers, technologists and managers
 - o At the level of society: as mutually enriching institutions and organizations
- Include practice Exercises and Case Studies will be taken up in Practice (tutorial) Sessions e.g. to discuss the conduct as an engineer or scientist etc.



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Text Books:

1. Human Values and Professional Ethics by R. R. Gaur, R. Sangal, G. P. Bagaria, Excel Books, New Delhi, 2010
2. Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. Jeevan Vidya: EkParichaya, A. Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.

Reference Books:

1. The Story of Stuff (Book).
2. The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi.
3. Small is Beautiful - E. F. Schumacher.

Course Outcome:

After completion of course, student should be able to

- To become more aware of themselves, and their surroundings (family, society, nature); they would become more responsible in life, and in handling problems with sustainable solutions, while keeping human relationships and human nature in mind.
- They would have better critical ability. They would also become sensitive to their commitment towards what they have understood (human values, human relationship and human society). It is hoped that they would be able to apply what they have learnt to their own self in different day-to-day settings in real life, at least a beginning would be made in this direction.